ERIK TAYLOR, PhD

AI Architect & ML Engineer | Agentic, Edge & Clinical AI Systems | Model Evaluation & Risk

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PhD engineer with 10+ years bringing applied AI from research and architecture to shipped products. Building agentic LLM applications and model-evaluation systems; deploying models to resource-constrained edge hardware — from browser-based LLMs to on-sensor vision. Led >$2M in NIH-funded computational imaging programs across cardiac, neuro-oncology, and traumatic brain injury, translating research into clinically validated systems with cross-functional teams spanning engineering, clinical, regulatory, and legal stakeholders. At Sony, drove end-to-end development of an on-sensor/on-device vision pipeline and orchestrated a high-stakes program recovery that rescued a flagship pre-market product. Evaluated frontier-model reasoning failures for Snorkel AI to inform RL reward signals and alignment strategy; shipped a privacy-first browser-LLM consumer health product from concept to production. Two decades of peer-reviewed research (50+ publications) ground a pragmatic engineering practice built on measurement, reliability, and safe deployment in regulated environments. Pursuing IAPP AIGP certification (Q3 2026).

PROFESSIONAL EXPERIENCE

AI Research Lead & ML Engineer *Oct 2024 – Present*

Taylor ML R&D | San Francisco, CA

- **Agentic AI Systems:** Architected apply-daemon (open-source) — an LLM-orchestrated agentic pipeline that ingests roles and triages through a multi-stage cascading-LLM workflow (extract → judge → evaluate), routing scored matches to a Slack human-in-the-loop layer where one reaction triggers deep-research resume synthesis. Cost-optimized via nano-tier model inference with on-demand escalation to frontier models.
- **Agentic Development Practice:** Build and ship production repositories primarily through agentic coding workflows — Claude Code (CLI and chat) for architecture, implementation, debugging, and iteration. Operate as the human-in-the-loop director of AI-assisted engineering for apply-daemon and the autonomous trading platform.
- **Technical Writing & Knowledge Leadership:** Author of “Digital Mind,” a technical series on frontier AI architectures (Diffusion Transformers, LLM reasoning, alignment, failure modes) that bridges academic research and practical implementation.

Founder & Lead ML Engineer *Sep 2025 – Feb 2026*

Napwise.ai | San Francisco, CA

- **On-Device Generative AI Product:** Shipped a consumer-facing, privacy-first infant sleep assistant running entirely in-browser. Architected a tiered inference engine (WebLLM/WebGPU, models down to Qwen 0.6B) with a hybrid RAG layer combining pre-cached JSON retrieval and LLM reasoning to minimize latency across diverse consumer hardware.
- **Edge Deployment & Cost Strategy:** Owned the transition to on-device AI to eliminate cloud inference costs and ensure data residency. Evaluated and selected efficient open-source models and RAG implementations to maintain performance across heterogeneous environments— BYO compute for LLMs.
- **Full-Stack Product Ownership:** Lead engineer from concept through production: product requirements, React frontend, scheduling engine, Cloudflare deployment, analytics, user testing, and roadmap iteration. Translated user insights into technical requirements to refine reliability and practical utility.

Senior Algorithm Engineer *Jul 2022 – Oct 2024*

Sony Electronics | San Jose, CA

- **On-Sensor & On-Device Computer Vision:** Led end-to-end development of a production two-stage vision pipeline for phones and VR headsets — a sub-1 MB CNN running on the image sensor that offloads heavier compute to the host device. Owned data curation, multi-GPU distributed training (DDP), INT8 quantization, and latency optimization (~30 FPS), securing a major B2B design win with a Fortune 500 customer.

- **Foundation-Model Roadmapping & AI Architecture:** Drove the technical roadmap for next-generation AI features by benchmarking SOTA vision and language foundation models — including a working prototype of open-vocabulary detection (OWL-ViT, CLIP) for promptable on-device perception — against performance, safety, and deployment constraints. Translated architectural trade-offs into product strategy.
- **Cross-Functional Program Recovery:** Orchestrated a high-stakes response to a sudden data policy change by bridging legal, business, and academic partners to co-develop a compliant dataset. Unblocked a next-generation AI hardware initiative and rescued a multi-month delivery timeline for a flagship pre-market product.
- **Model Validation & Safety Criteria:** Defined evaluation protocols for edge AI products targeting regulated markets, establishing internal standards for performance, bias, and deployment constraints aligned with responsible AI principles.

AI Evaluation Specialist · Contract *Feb 2025 – Aug 2025*
Snorkel AI | Remote

- **Frontier Model Risk & Reliability:** Authored advanced mathematical and graduate-level test cases designed to expose reasoning failure modes in frontier LLMs. Findings directly informed RL reward signals and alignment strategy for next-generation models and produced technical insights for production-readiness risk management.

Research Assistant Professor, Medical Imaging AI *Apr 2019 – May 2022*
University of New Mexico — Medical Physics & Radiology | Albuquerque, NM

- **Technical Program Leadership:** Led the AI strategy and computational architecture for a multi-disciplinary center with >$2M in active grant funding. Governed imaging data pipelines and model development across cardiac, neuro-oncology, and traumatic brain injury studies.
- **Biomedical Data Architecture (Journal Cover Feature):** Rescued a stalled traumatic brain injury study by architecting a custom deep learning framework (U-Net variant) capable of processing noisy, unstandardized biomedical data. Validated the architecture against baselines, earning a cover feature in the Journal of Neurotrauma (2023).
- **Cross-Functional Clinical Integration:** Bridged engineering researchers and clinical stakeholders (Radiology, Neurology, Oncology) to translate AI research into actionable clinical validation studies. Deployed HIPAA-compliant data handling, IRB-governed protocols, and clinical data governance across multi-site imaging studies.

NIH Research Fellow, Cardiovascular MRI *Jul 2016 – Mar 2019*
Boston University | Boston, MA

- **Translational Imaging Research:** Developed and validated MRI acquisition and analysis methods for cardiovascular disease, combining physics-informed feature engineering with deep learning, and establishing a framework for choosing deep learning vs. classical methods by efficiency and clinical fit.

SELECTED PROJECTS

- **Browser-Based Edge Inference (napwise.ai).** React + WebLLM application for offline-capable generative AI. Implemented client-side sparse vector search (TF-IDF/cosine similarity) running purely in-browser to rank documents without external APIs.
- **Autonomous Portfolio Manager.** LightGBM proposes tickers/themes; an LLM-orchestrated agent layer reasons over tools (financials, web search, SEC filings) to generate investment rationales; a Python/LLM portfolio manager executes trades via Alpaca with risk guardrails and paper/live modes, plus a Streamlit dashboard for live monitoring and human-in-the-loop trade review. Full CI/CD on GitHub Actions; migrating orchestration to a LangGraph-style graph layer (in progress).
- **Transformer Architecture Series (eriktaylor.github.io).** Author of an open-source engineering series providing reference implementations, mathematical teardowns, and optimized codebases for ViT, CLIP, Diffusion Transformers, DINOv2, and LLM failure mechanisms.

TECHNICAL SKILLS

Agentic AI & LLM Applications: LLM application development (RAG, agents, tool use, function calling) · Claude / Claude Code (daily use) · Anthropic API (via OpenRouter) · LangChain · LangGraph · LlamaIndex · autonomous agent orchestration · Model Context Protocol (MCP) · multi-modal AI · prompt engineering

Model Evaluation & AI Safety: Frontier-model reasoning failure analysis · evaluation harness design · model risk and validation · benchmarking (accuracy / latency / safety / cost trade-offs) · red-teaming · alignment-informed test design

Computer Vision & Edge AI: Detection, depth estimation, segmentation · open-vocabulary detection (OWL-ViT, CLIP) · self-supervised vision (DINO/DINOv2) · YOLO family · on-device / on-sensor inference · split/cascaded inference · quantization (INT8/INT4) · knowledge distillation · ONNX / WASM · browser-based inference (WebLLM, WebGPU) · GPU profiling

Frameworks & Languages: Python (expert) · PyTorch · TensorFlow/Keras · Hugging Face Transformers · scikit-learn · LightGBM/XGBoost · FAISS · JavaScript · SQL · C++ (working) · OpenCV

MLOps & Infrastructure: AWS · GCP · Docker · CI/CD (GitHub Actions) · Cloudflare (Pages, Workers) · distributed training (DDP, FSDP) · experiment tracking (W&B) · Linux/Ubuntu

Domain Knowledge: Medical imaging (MRI, PET, CT) · physics-informed ML · clinical data governance · translational research workflows · edge hardware constraints (memory, latency, power)

STRATEGIC CAPABILITIES

AI Product Strategy & Risk Management: Directing model evaluation and vendor selection · foundation model benchmarking · agentic and algorithmic stress-testing · defining model validation constraints · driving edge vs. cloud deployment strategy.

Cross-Functional Program Leadership: Technical program management and full product lifecycle ownership · stakeholder management across clinical, legal, regulatory, and engineering teams · external partner and vendor coordination · matrixed team leadership.

Clinical AI & Regulatory Awareness: FDA SaMD landscape (PCCP, GMLP, IEC 62304 — actively developing) · EU AI Act high-risk classification · NIST AI RMF · ISO 42001 · HIPAA · IRB-governed research protocols · pursuing IAPP AIGP certification (Q3 2026).

EDUCATION & RESEARCH TRAINING

PhD, Biomedical Engineering — Brown University
Doctoral research in theranostic technologies, integrating advanced medical imaging diagnostics with targeted therapeutics.

NIH Research Fellow, Cardiovascular & Brain MRI — Boston University
Federally funded fellowship developing and evaluating deep learning frameworks for translational medical imaging.

BS, Biomedical Engineering — University of Texas at Austin
Graduate Coursework, Data Science — Harvard University

SELECTED PUBLICATIONS

Selected from 50+ peer-reviewed publications. Full list available on request.

Divani AA, Salazar P, Ikram HA, **Taylor E**, Wilson CM, et al. Non-invasive vagus nerve stimulation improves brain lesion volume and neurobehavioral outcomes in a rat model of traumatic brain injury. *Journal of Neurotrauma*. 2023; 40(13–14): 1481–1494. **★ Journal Cover Feature**

Taylor EN, Huang N, Lin S, Mortazavi F, Wedeen VJ, et al. Lipid and smooth muscle architectural pathology in the rabbit atherosclerotic vessel wall using Q-space cardiovascular magnetic resonance. *Journal of Cardiovascular Magnetic Resonance*. 2022; 24(1): 74.

Taylor EN, Ding Y, Zhu S, Cheah E, et al. Association between tumor architecture derived from generalized Q-space MRI and survival in glioblastoma. *Oncotarget*. 2017; 8(26): 16296.